

Freight Rate Prediction: Technical Report

Validation Approach, Model Selection, and December Prediction Report - Misiker Kassahun Zewde

1. Executive Summary

This report describes the end-to-end modeling approach for predicting freight load rates. The final model is a HistGradientBoostingRegressor trained on a log-transformed target, validated against a time-based holdout, and refit on the full training corpus before generating predictions for 12,000 validation loads and a fixed lane across December 2025.

2. Data Exploration & Quality Issues

The development data spans January through October 2025 (48,000 loads). Key findings from exploration:

- **Target distribution:** posted_rate is right-skewed, with a long tail of high-rate loads.

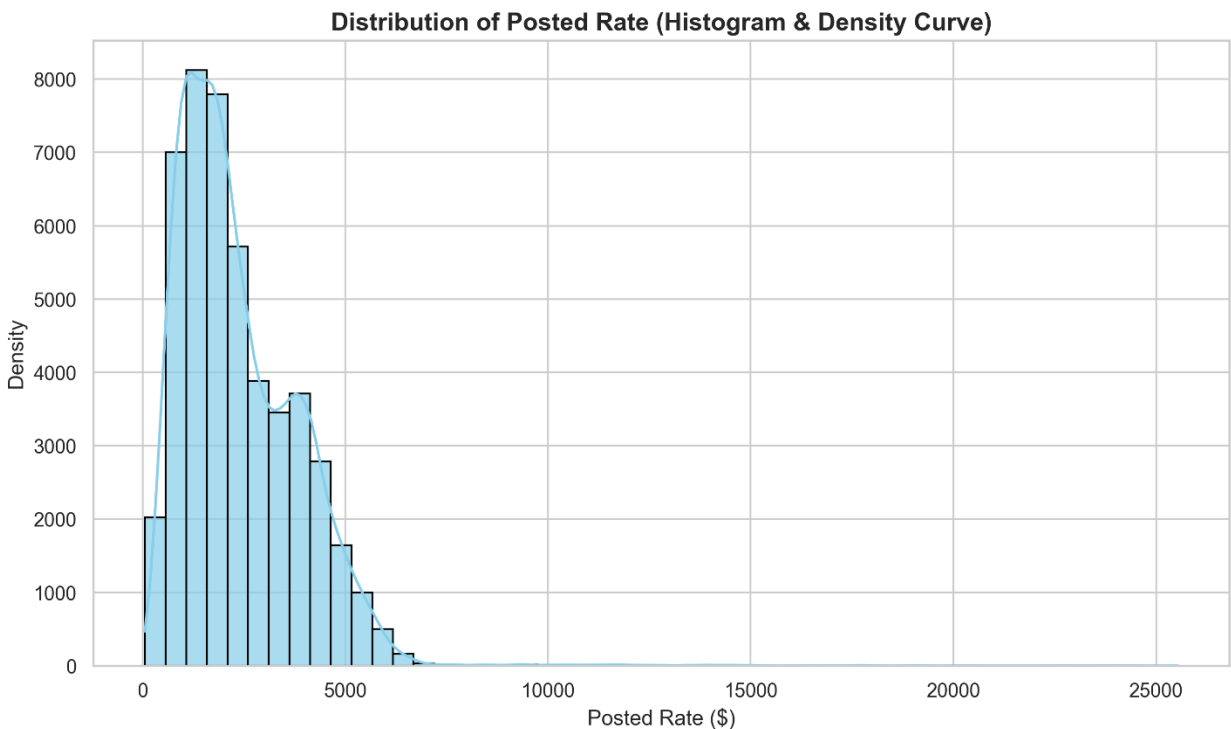


Figure 1. Histogram distribution and density curve of posted_rate.

- **Outliers:** A small cluster of loads exhibits rates 3-7× the typical per-mile rate, suggesting data-entry anomalies or premium expedited shipments.
- **Missingness:** weight and market_index contain nulls.
 - weight: 300 missing values (~0.6%)

- `market_index`: 374 missing values (~0.8%)
- **Sign errors:** Some weight values are negative, indicating entry mistakes.

Solution:

- Negative weights were corrected by taking the absolute value.
- Missing weight and market index values were imputed using the training-split median only, preventing validation-set leakage.
- No rows were dropped; all usable signal was preserved.

3. Data Split and Validation Approach

`train_test.csv` covers loads from January through October 2025, but the actual job is to predict loads in `validation.csv` (which sit in the Nov–Dec window) and a full month of December for one fixed lane. That's a forecasting problem, predicting rates for time periods the model hasn't seen, which is not a problem where rows can be shuffled and split randomly.

A random k-fold or random train/test split would let the model see a load from October 30th right next to a load from early October, and it would report an optimistically low validation score that doesn't reflect how the model performs when it has to extrapolate weeks or months forward. That mismatch is exactly the gap between `train_test.csv` and `validation.csv`,

Instead, a time-based split was used:

- Training set: all loads dated before September 1, 2025 (about 38,477 rows).
- Holdout set: all loads from September 1 through October 31, 2025 (about 9,523 rows), held out entirely from training.

This two-month gap mirrors the deployment gap between `train_test.csv` and `validation.csv`, yielding an honest estimate of generalization to unseen future dates. The model is trained only on the past and evaluated on a period it never saw, the same way it will later be asked to predict November, December, and the fixed lane. Any imputation values (like median weight and median market index) were also computed on the training portion only and then reused on the holdout set, so no information from the holdout period leaked backward into training.

4. Feature Engineering

Features were constructed in three layers:

1. **Geographic:** City latitude/longitude lookups were built from the training data and used to backfill any missing coordinates.
2. **Market:** `market_index` and `quote_signal` were filled with training-derived medians when absent.

3. **Temporal:** Cyclical encodings (month_sin, month_cos, doy_sin, doy_cos), calendar indicators (day_of_week, is_weekend, week_of_year), and raw month/day-of-year were added to let the model learn seasonal patterns without imposing a linear time trend.

Categorical columns (pickup, delivery, equipment) were cast to pd.Categorical with levels fixed from the training data, enabling native categorical support in HGBR.

5. Model Selection & Rationale

Gradient Boosted Decision Trees (GBDTs) were selected over deep neural architectures due to their performance on tabular data and robust handling of categorical features. For the model itself, I used a HistGradientBoostingRegressor (HGBR) trained with an L1 (absolute error) loss, this choice prevents a small group of rate outliers (which are much higher than expected for their mileage) from skewing the model. Because HGBR's built-in early stopping carves a *random* validation fraction from the training data (not the chronological holdout of September-October window), it was disabled. A manual warm-start loop was implemented and checked the true holdout MAE every 20 boosting rounds, stopping it if it hasn't improved for four consecutive checkpoints. The model was then refit from scratch on the entire train_test.csv at the best round (680 iterations) to maximize data usage before deployment. To validate this selection, LightGBM was also trained and evaluated under identical conditions. While LightGBM achieved nearly identical performance, HGBR demonstrated a slight edge in predictive accuracy.

Key hyperparameters:

- loss="absolute_error": L1 objective is robust to the outlier-rate loads discovered during EDA.
- learning_rate=0.05, max_leaf_nodes=63, min_samples_leaf=30, l2_regularization=1.0, moderate regularization to prevent overfitting on the time-based holdout.
- categorical_features="from_dtype": native handling of city and equipment categories without one-hot explosion.

Baseline: A linear regression model was trained on the same features on one-hot encoded categoricals + numeric features for sanity-check purposes.

Metric	Baseline (Linear Regression)	HGBR (Final Model)
MAE (\$)	433.29	110.43
RMSE (\$)	869.69	635.36
MAPE	20.64%	4.76%
R ²	0.675	0.827

The HGBR model lowered the error from about \$433 to about \$110 and reduced the MAPE from 20.6% to 4.8%. This improvement was measured using data that the HGBR model was not trained on. The training curve below shows holdout MAE (log scale) flattening out around the mid-100s of boosting rounds, with the best checkpoint landing at iteration 680 before further rounds stopped helping.

Feature importance (permutation on holdout):

- distance dominates (0.704), as expected for freight pricing.
- equipment (0.036) and weight (0.010) are secondary drivers.
- Origin/destination cities and market signals contribute modestly.

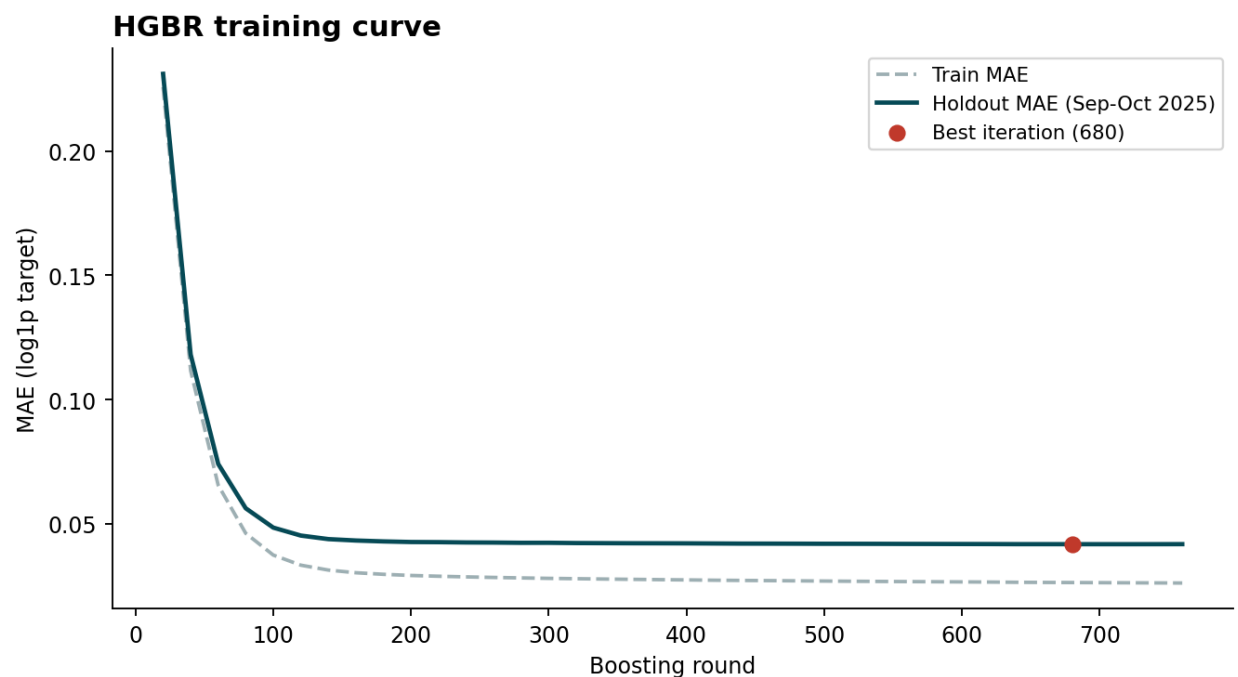


Figure 2. HGBR training curve. Holdout MAE plateaus around iteration 680, indicating adequate regularization and no severe overfitting.

6. December 2025 Fixed-Lane Forecast

The chart below was produced by the provided score.py and shows the predicted rate for a fixed lane (Lexington → Fort Wayne, 360 mi, Dry Van, 32,000 lb) across every day of December 2025. The rate hovers near \$837-\$839 for most of the month, with modest dips around weekends and a slightly larger step down during the final holiday week (December 27-31), to roughly \$833-\$834. This pattern is consistent with typical freight market seasonality.

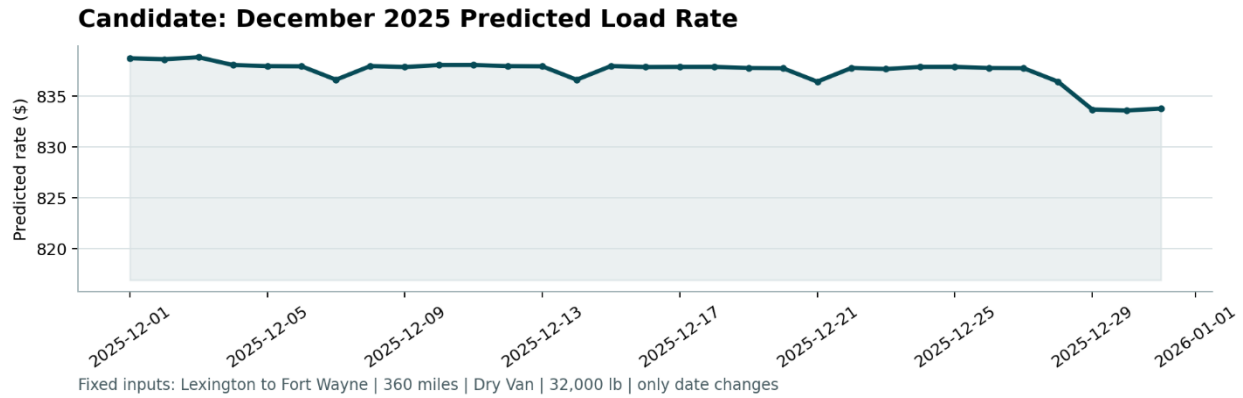


Figure 3. *candidate_december.png*: predicted rate for the fixed Lexington-Fort Wayne Lane across every day of December 2025.

Since pickup, delivery, distance, equipment, and weight are all held fixed, that movement is coming entirely from the date-based features (day of week, day of year, seasonality encodings) and reflects the small late-December effect the model picked up from historical patterns rather than any change in the lane itself.

7. Conclusion

The solution follows forecasting best practices: a time-respecting validation split, leakage-free preprocessing, robust feature engineering, and a tree-based model with an L1 objective chosen specifically to handle rate outliers. The large margin over the linear baseline (MAE reduced by 75%) and the stable training curve indicate a well-generalized model ready for the November–December prediction window.